

Edge-side AI Recognition, Accurate and Fast Settlement



AE510 Smart Kit

· Edge Computing

· 14 TOPS AI

· 4-Camera Vision

· Smart Lock

1. Product Overview

AI vending machine retrofit kit with edge-side AI recognition for accurate, fast automated checkout.

Key Features:

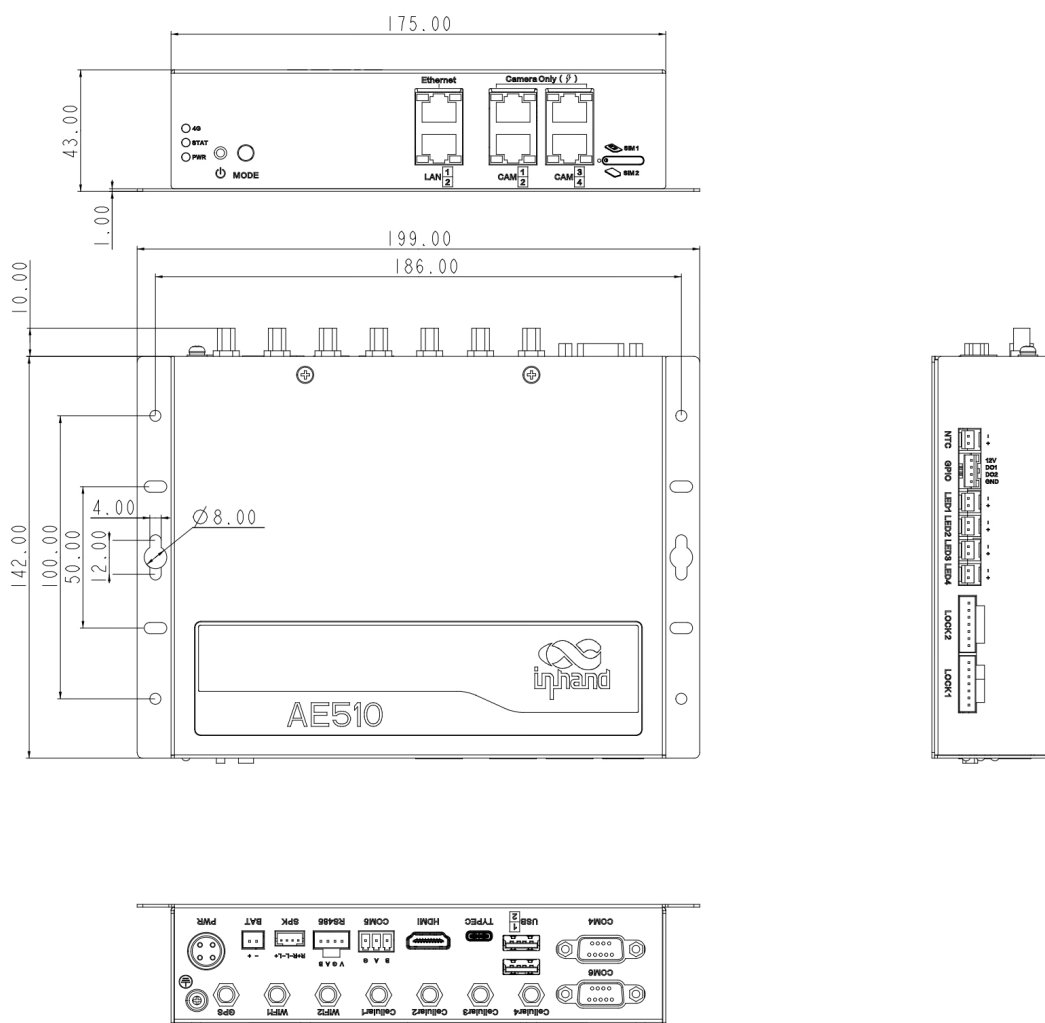
- **Edge-side AI:** 14 TOPS AI computing power with octa-core processor for real-time recognition
- **Vision System:** 4 × AI cameras with advanced image recognition for accurate item capture
- **Flexible Retrofit:** Highly compatible upgrade solution for traditional vending machines
- **All-in-One Kit:** Integrated smart lock, backup battery, and voice speaker
- **Open Platform:** Android 12 OS supporting custom application development

Core Technical Specifications

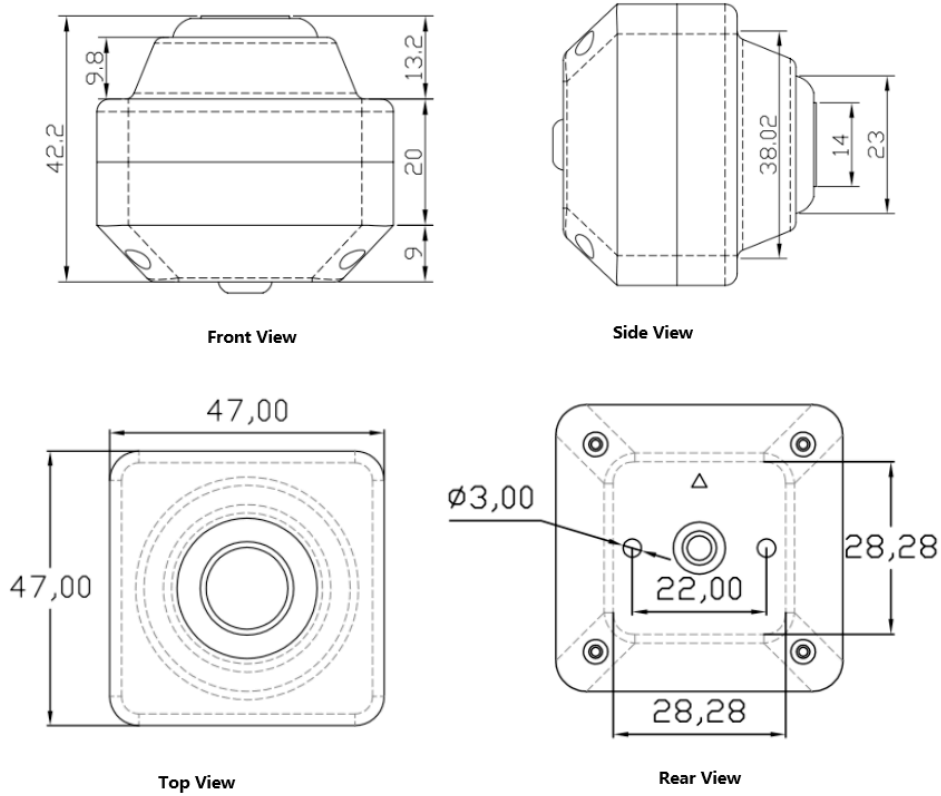
Item	Specification
AI Computing Power	14 TOPS
OS	Android 12
Edge Recognition	Real-time AI image recognition
Camera Protocol	ONVIF, H.264/H.265, 1080P @ 25 fps
Lock Control	Smart electronic lock, LED status indicator
Voice Output	5 W speaker support
CPU	Octa-core, up to 2.7 GHz
RAM / Storage	8 GB / 128 GB
Ethernet	4 × 10/100 Mbps (PoE) + 2 × 10/100 Mbps

Item	Specification
Dimensions (W × D × H)	199 × 152 × 43 mm
Power Input	12 V DC
Operating Temperature	-10 ~ 60 °C

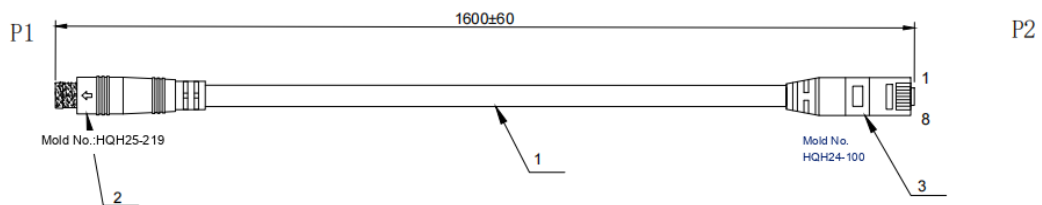
2. Product Dimensions



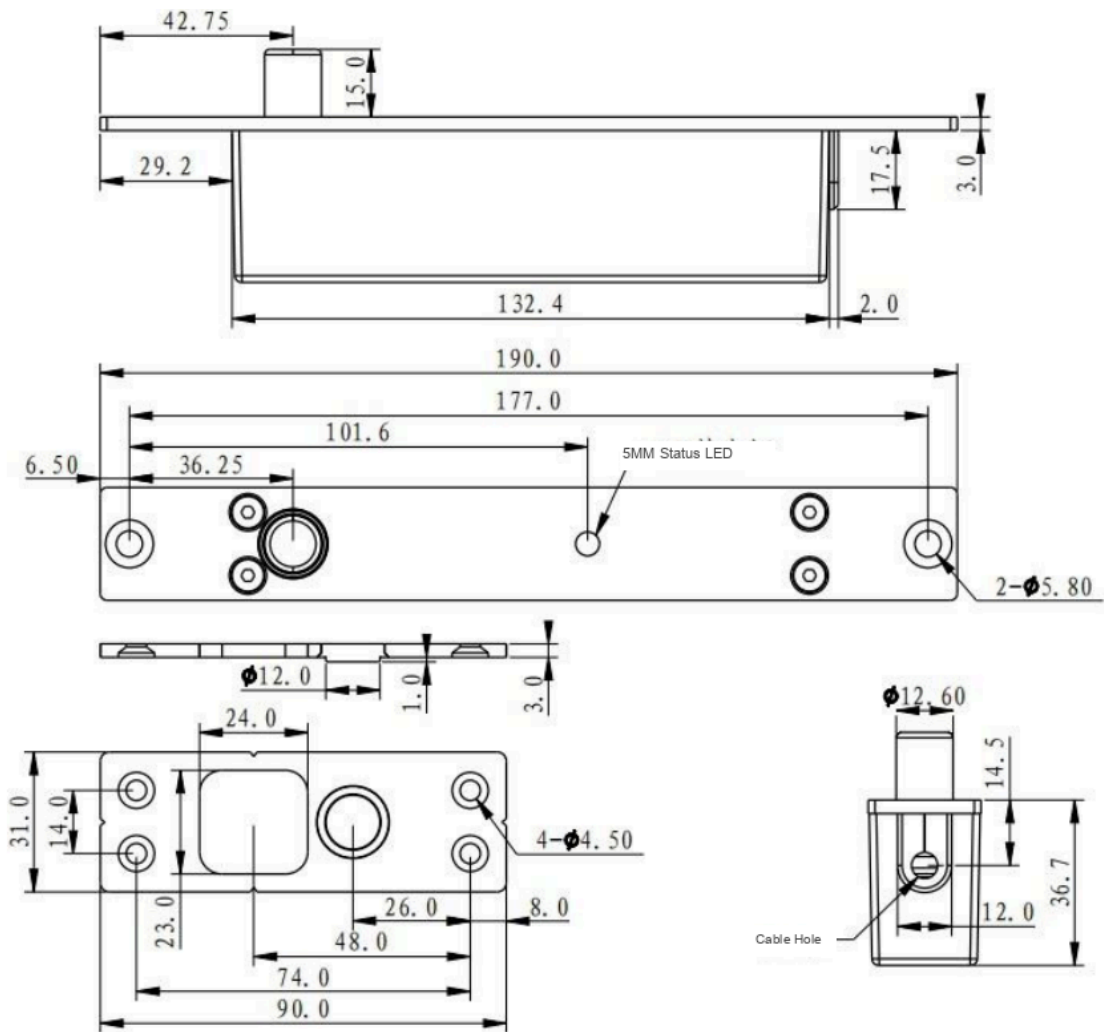
AE510



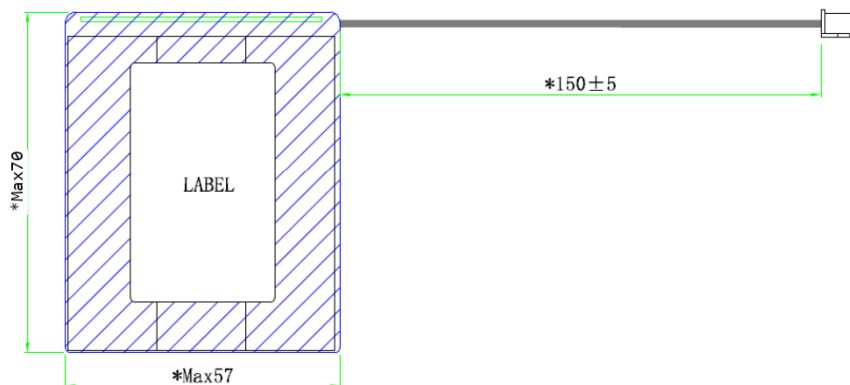
AI Camera



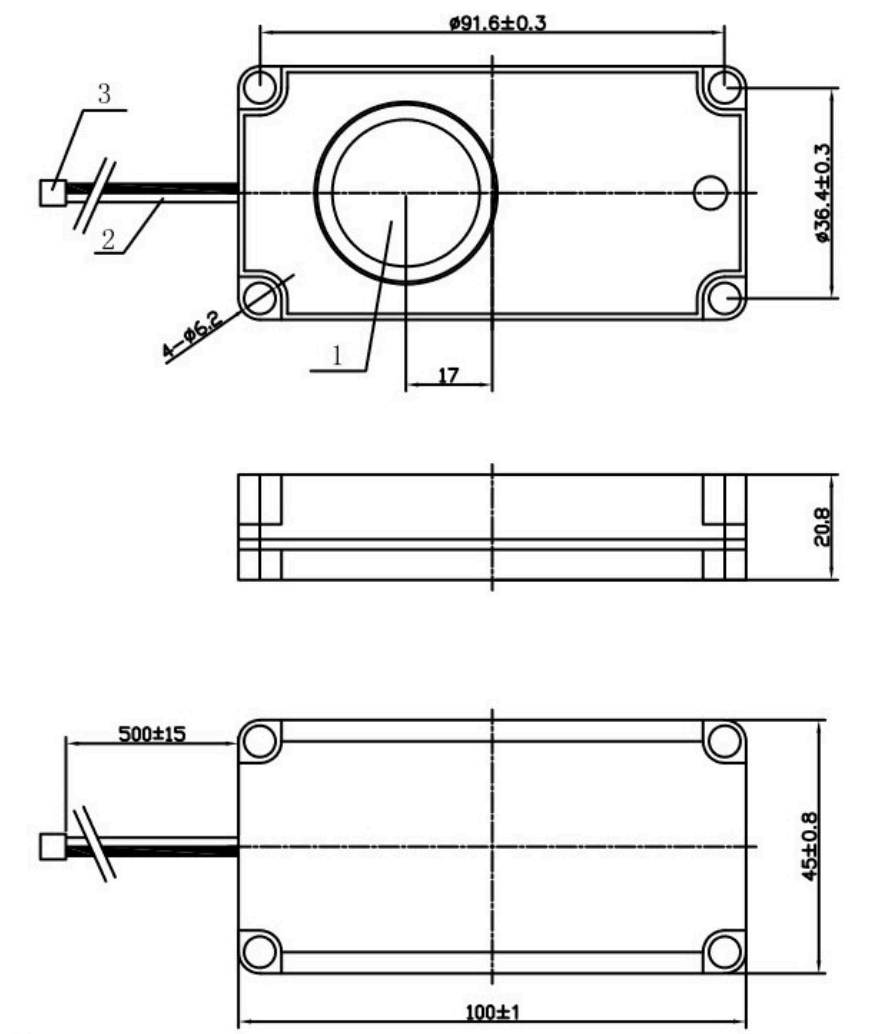
AI Camera Cable



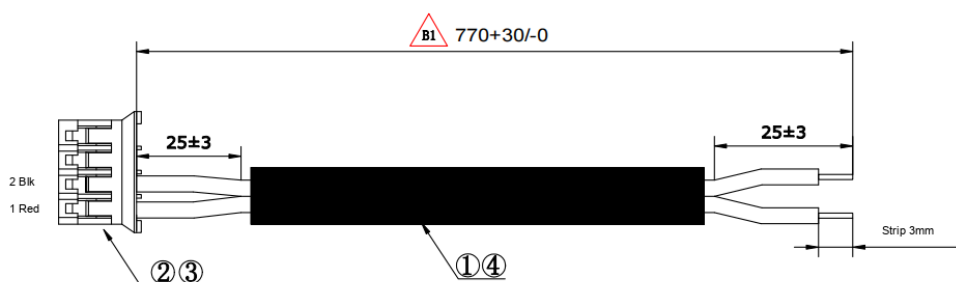
LOCK



Battery



Speaker



Speaker Cable

Note:

1. All dimensions are in millimeters (mm).
2. All dimensions are approximate and **for reference only**.
3. Illustrated dimensions **must not be used for production**.
4. Dimensions are subject to component and manufacturing tolerances.
5. Dimensions are subject to change without notice.

Interface Definition

AE510 Interfaces

Interface	Quantity	Specification
Ethernet	4	10/100 Mbps RJ45 with PoE (for IP cameras)
Ethernet	2	10/100 Mbps RJ45
SIM	2	Nano SIM Slot
RS232	2	DB9
RS485	2	15EDGRC-3.81 / HX25035-4WAP2
USB Type-A	2	—
USB Type-C	1	—
HDMI	1	—
Lock Interface	2	HX25035-7Y
Battery Interface	1	HX25003-2WAP2
Speaker Interface	1	Supports 5 W speaker
DO (Digital Output)	2	S04B-XASK-1
NTC	1	S02B-XARK-1
LED	4	S02B-XASK-1
Antenna (4G)	2	SMA
Antenna (Wi-Fi)	2	RP-SMA

3. Hardware Specifications

Category / Parameter	Specification
Processor	
CPU	Octa-core, up to 2.7 GHz
AI Computing Power	14 TOPS
RAM	8 GB
Storage	128 GB
OS	Android 12
Interface	
SIM	2 × Nano SIM Slot
Ethernet	4 × 10/100 Mbps RJ45 (PoE for IP cameras) + 2 × 10/100 Mbps RJ45
Serial Port	2 × RS232 (DB9), 1 × RS485 (15EDGRC-3.81), 1 × RS485 (HX25035-4WAP2)
USB	2 × Type-A, 1 × Type-C
HDMI	1 × HDMI
Lock Interface	2 × HX25035-7Y
Speaker Interface	1 × SPK, supports 5 W speaker
Battery Interface	1 × HX25003-2WAP2
DO (Digital Output)	2 × S04B-XASK-1
NTC	1 × S02B-XARK-1
LED	4 × S02B-XASK-1

Category / Parameter	Specification
Antenna	
4G Antenna	2 × SMA
Wi-Fi Antenna	2 × RP-SMA
Buttons & Indicators	
Buttons	1 × Power Button, 1 × Mode Button
Indicators	1 × Power, 1 × Status, 1 × 3G/4G
AI Camera	
Quantity	4
Lens	F:2.0, FOV D: 140°, H: 115°, V: 65°, Focal length 3.0 mm
Image Sensor	1/2.8" CMOS
Max Resolution	1920 × 1080
Max Frame Rate	25 fps
Low Illumination	Supported
Wide Dynamic Range	Supported
White Balance	Auto
3D Noise Reduction	Supported
Encoding Format	H.264 / H.265
Protocol	ONVIF
Power Input	12 V DC
Power Consumption	≤ 1.8 W

Category / Parameter	Specification
Operating Temperature	-10 ~ 50 °C
Smart Lock	
Working Voltage	12 V DC $\pm$ 10%
Working Current	Start 1.2 A, Steady 0.13 A
Panel Size	190 × 25 × 3 mm
Lock Dimensions	190 × 25 × 37 mm
LED Indicator	Green (unlock), Red (lock)
Mechanical Life	$\geq$ 300,000 cycles, failure rate < 3%
Speaker	
Dimensions	Φ 100 × 45 mm
Magnet	Φ 12.5 × 4.0 mm Nd-Fe-B
Nominal Power	3.0 W
Max Power	5.0 W
Backup Battery	
Dimensions	70 × 57 × 20 mm
Typical Capacity	2000 mAh
Nominal Voltage	11.1 ~ 12.6 V
Max Discharge Current	2 A
Cycle Life	$\geq$ 300 cycles @ 80% capacity retention
Operating Temperature	Charging: 0 ~ 45 °C, Discharging: -20 ~ 60 °C

Category / Parameter	Specification
Protection	Overcharge, overdischarge, overcurrent, short circuit
Mechanical	
Dimensions (W × D × H)	199 × 152 × 43 mm (including installation parts)
Housing	Metal
Cooling	Fanless
Protection Rating	IP40
Environmental	
Operating Temperature	-10 ~ 60 °C
Storage Temperature	-40 ~ 85 °C
Humidity	5 ~ 95% (non-condensing)
Compliance	
EMC	ESD / EFT / Surge: Level 2

4. Software Specifications

Category / Parameter	Specification
Operating System	
OS	Android 12
Camera & Video	
Video Encoding	H.264 / H.265
Image Resolution	1080P / 720P / 360P / D1 @ 25 fps

Category / Parameter	Specification
Transmission Protocol	ONVIF
Image Processing	Auto white balance, 3D noise reduction, wide dynamic range
Low Illumination	Supported
Edge Computing	
AI Computing Power	14 TOPS
Recognition	Edge-side AI image recognition
Development Environment	Android application development support
Network	
Cellular	4G LTE
Wi-Fi	2.4 GHz
Ethernet	4 × 10/100 Mbps PoE + 2 × 10/100 Mbps

5. Ordering Information

Model List

Code	Model	Description
GPRO310071	AE510 Dynamic Vision Kit	AI Vending Machine single door cabinet, AE510 version dynamic vision kit, local recognition solution. Includes AE510 controller, AI camera × 4, smart lock, speaker, backup battery, etc.
SLCD000007	InTouch-ZXD08	8-inch widescreen display
SMDM120002	InPOS-IM30	POS Payment module

Note: Order network cables according to installation length.

6. Contact Us

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